

Did we ask these questions first?

Every question on the four OPF First Term 2026 papers, checked against all 65 practice papers · 7 September 2026

The short version. Yes, and far more often than I expected. I pulled the text out of all 65 practice papers we have built since April and checked every single question on the school's four papers against them. Of 82 question slots on the school papers (counting each question and its OR alternative separately), **74 had been set before** in one of our papers. Eight of those were near-identical repeats, including one chemistry question that matched down to the exact numbers. Only eight were genuinely new.

The standout: the school's Chemistry Section B Q8 asked for the enthalpy of $\text{H}_2 + \text{Cl}_2$ using bond energies $\text{H-H} = 436$, $\text{Cl-Cl} = 243$, $\text{H-Cl} = 432$ kJ/mol. Our chemistry exam-05 and again exam-13 asked that exact question with those exact three numbers. She had done it twice.

8

near-verbatim repeats

66

same topic, set before

8

genuinely new

The eight near-verbatim repeats

Subject	School question	Our version	Where
Chemistry	Q8: enthalpy of $\text{H}_2 + \text{Cl}_2$ from $\text{H-H} = 436$, $\text{Cl-Cl} = 243$, $\text{H-Cl} = 432$	"Calculate ΔH for $\text{H}_2 + \text{Cl}_2 \rightarrow 2\text{HCl}$ using bond energies: $\text{H-H} = 436$, $\text{Cl-Cl} = 243$, $\text{H-Cl} = 432$ kJ/mol." Identical reaction, identical data.	exam-05 3(ii)(b) exam-13 3(iv)(a)
Chemistry	Q9 OR: how does temperature affect the cobalt chloride equilibrium	"Explain the effect of temperature on the hydrated cobalt(II) chloride equilibrium. Write the equation and describe the colour changes."	exam-13 3(iv) OR also exam-06, exam-16
Maths	Q4: cubical container of volume $(x^3 - 6x^2y + 12xy^2 - 8y^3)$ m ³ ; find side, base area, total surface area, painting cost at Rs 50/m ²	"The volume of a cubical oil tank is $(125x^3 - 150x^2 + 60x - 8)$ m ³ . Find (a) the side (b) the total surface area (c) the cost of painting it from outside at Rs 32/m ² ." Same perfect-cube factorization, same part structure.	exam-13 3(iii)
Physics	Q v: liquid of three times the density of mercury — height of the barometer column	"If the same barometer is filled instead with a liquid of density 3,400 kg/m ³ , calculate the height of the column... State how the height depends on the density of the liquid."	exam-14 2(iii)
Physics	Q ix: differentiate theory and law with 2 examples each	"Define and explain hypothesis, theory, and law with two examples of each. What is the difference between a theory and a law?"	exam-02 (Section C)
Biology	Q ii OR: main features of the Watson and Crick model of DNA	"State THREE main points of the Watson and Crick model of DNA structure."	exam-07 2(x) also exam-08, exam-12
Biology	Q iv OR: briefly explain the types of RNA	"Name the three types of RNA and state the specific role of each in protein synthesis." This was the OR alternative to the Watson-Crick question in the same row of exam-07 — the school used the identical pairing, split over two rows.	exam-07 2(x) OR
Biology	Q i OR: cause, vector and symptoms of dengue fever	"What is Dengue fever? State its viral cause, its mosquito vector, and the recommended method of prevention."	exam-10 2(i) OR

The eight that were genuinely new

Subject	School question	Closest thing we did set
Chemistry	Q3: Why is an atom electrically neutral?	Nothing. Subatomic particle charges were covered but never this framing.
Chemistry	Q5: Why do transition elements show variable oxidation state?	Nothing. The word "variable" appears in no practice paper.
Chemistry	Q6 OR: balanced equation for burning hydrogen to produce water	Balancing was drilled heavily, but never this reaction.
Chemistry	Q7: derive the formula of aluminium sulfate	Ionic-formula writing appears, but never a derivation of this compound.
Physics	Q ii OR: 600 N on Earth, weight on the Moon where $g = 1.62$ m/s ²	Nothing. $W = mg$ was used, but no Moon-gravity problem in any paper.
Biology	Q ii: structural conformation of proteins	Proteins were covered as bioelements/macromolecules, but primary/secondary/tertiary conformation was never asked.
Biology	Q iii: why is it impossible to eradicate malaria?	Malaria appears in five papers (biological method, Ronald Ross, vector), but never the eradication question.
Maths	Q v: first polynomial $x^2 - x - 6$, HCF $(x - 3)$, LCM $x^3 + 11x^2 + 12x - 12$; find the other polynomial	HCF and LCM were drilled in nine papers, but never the reverse "product = HCF $\times$ LCM" application.

Four worth calling out as near-misses that still landed

- **Physics Q iv OR (why a helmet is safe, with a mathematical relation).** My first pass called this a miss because no paper contains the word "helmet". It is not a miss: exam-11 2(vi) set "an airbag brings a 70 kg crash-test dummy from 15 m/s to rest in 0.2 s, calculate

the change in momentum and the average force". Same impulse relation, different object.

- **Biology Q v OR (name the main types of plastids).** Exam-10 covered plastids across two separate questions, chloroplast structure and chromoplasts vs leucoplasts, but never as one "name them all" question.
- **Maths Q iv (show three points form a right-angled triangle).** We drilled the technique repeatedly with other shapes: "show the triangle is isosceles" (exam-11), "prove these are the vertices of a square" (exam-11), rectangle and rhombus proofs (exam-11, exam-12). Never a right angle specifically.
- **Chemistry C2 OR (mass of 3×10^{10} molecules of oxygen).** Avogadro and mole-to-mass conversions appear in six papers, including "calculate the total number of atoms in 0.5 mole of nitrogen gas" (exam-14). Same machinery, run backwards.

Per-subject tally

Subject	Question slots	Near-verbatim	Same topic, set before	New	Covered
Chemistry	22	2	16	4	82%
Physics	22	2	19	1	95%
Biology	22	3	17	2	91%
Mathematics	16	1	14	1	94%
Total	82	8	66	8	90%

A "question slot" counts the main question and its OR alternative separately, since both were printed and either could be attempted. Section A (the objective sheets) is not included because those papers were never photographed. Method: pdftotext -layout over all 65 practice papers, keyword search per school question, then manual reading of the actual wording for every candidate match. Keyword hits alone were not treated as matches.

What this changes

- **Chemistry is the weakest-covered of the four** at 82%, and all four of its gaps are short recall questions rather than hard ones: atom neutrality, variable oxidation state of transition metals, one balanced equation, one formula derivation. Cheap to close in the next chemistry paper.
- **The saturation warnings were wrong to act on.** Our trackers flagged hypothesis/theory/law as over-drilled in physics and Ch 8-9 as exhausted in chemistry, and we had been avoiding both. The school set all of them. Repetition of core material is not waste when the school examines the same core.
- **Exam-13 in both chemistry and maths did unusual work.** Chemistry exam-13 alone hit three of the school's questions (bond energy, cobalt chloride, oxidation state of N in HNO_3); maths exam-13 produced the cubical-container match. Both were built in late July under the coverage-discipline rule.